

EDUCATION

Yale University

Ph.D. in Computer Science

New Haven, CT

In progress

- Research areas: statistical learning theory, machine learning theory

University of Pennsylvania

M.S.E. in Computer Science, GPA: 4.00/4.00

Philadelphia, PA

May 2019

- Submatriculant: Completed Bachelor's and Master's degrees concurrently
- Thesis: "Noisy Labels in Multiclass Classification", advised by Dr. Shivani Agarwal

University of Pennsylvania

B.A. in Mathematics and in Computer Science, minor in Statistics, GPA: 3.84/4.00

Philadelphia, PA

May 2019

- Summa Cum Laude

SELECTED SCHOLARSHIPS AND AWARDS

- **Greenberg Fellowship**, Yale Graduate School of Arts and Science (GSAS) 2024
Named Yale fellowships recognize graduate students' academic excellence. They are also a reflection of GSAS alumni and friends' commitment to support the individual student.
- **Distinguished Teaching Award**, Yale Computer Science 2023
One (1) graduate student recognized for outstanding teaching during the 2022-2023 academic year.
- **P.E.O. Scholar Award**, Nominee 2023
Nominated by Chapter AA of P.E.O. International for the P.E.O. Scholar Award for the 2023-2024 cycle.
- **Quad Fellowship by Schmidt Futures**, Finalist 2022
Finalist (<5% of applicants from U.S.A., India, Australia, and Japan) for the Quad Fellowship 2022-2023 cycle.
- **Graduate Fellowship for STEM Diversity** 2022
Formerly the National Physical Science Consortium Fellowship (NPSC), the GFSD is a national fellowship which provides 6 years of support and \$20,000 stipend. Supported by the National Security Agency (NSA).
- **Ford Foundation Predoctoral Fellowship Competition**, Honorable Mention 2022
Accorded honorable mention status in the 2022 Ford Foundation Fellowship Programs competition administered by the National Academies of Sciences, Engineering, and Medicine.
- **Department of Defense SMART Scholarship**, Declined 2020
Offered up to 5 years of support in addition to a \$38,000 living stipend. Awarded to the top 20% of Ph.D. applicants.
- **Twitter GHC Fellowship** 2018
A merit-based fellowship awarded to attend the 2018 Grace Hopper Celebration and a full-time offer of employment at Twitter.

WORKING PAPERS

Note: For many papers, authors are listed in alphabetical order.

1. **J. H. Lee**, N. Jali, I. Han, A. Zandieh, J. Mei, J. Baek, "Streaming Attention Approximation with Page-Aligned Memory Management" (in submission)
2. N. Jali, **J. H. Lee**, A. Nayak, C. Lin, J. Baek, "AnchorRank: Training-Free Anchor based Plan-and-Execute Decoding in DLMs" (in submission)

PUBLICATIONS

- [1] J. N. Kaur, **J. H. Lee**, and M. Zampetakis, “Long-term risks of risk-based allocation”, in *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*, forthcoming, Dec. 2026.
- [2] V. Kher, **J. Lee**, A. Mehrotra, and M. Zampetakis, “Learning-enabled estimation: Tight characterizations under sample selection biases”, in *The Twenty-Seventh ACM Conference on Economics and Computation (EC)*, Jul. 2026.
- [3] **J. Lee**, A. Mehrotra, and M. Zampetakis, “Smoothed analysis of learning from positive samples”, in *58th ACM Symposium on Theory of Computing (STOC)*, Jun. 2026.
- [4] Z. Chen, C. Zhao, K. C. Mo, Y. Jiang, **J. H. Lee**, K. C. Mahajan, N. Jiang, K. Ren, C. Li, and W.-Y. Yang, “Massive memorization with hundreds of trillions of parameters for sequential transducer generative recommenders”, in *The Fourteenth International Conference on Learning Representations (ICLR)*, Apr. 2026.
- [5] **J. Lee**, B. Saglam, S. Pougkakiotis, A. Karbasi, and D. Kalogieras, “Risk-averse constrained reinforcement learning with optimized certainty equivalents”, in *Advances in Neural Information Processing Systems (NeurIPS)*, Dec. 2025.
- [6] **J. Lee**, A. Mehrotra, and M. Zampetakis, “Efficient statistics with unknown truncation: Polynomial time algorithms, beyond gaussians”, in *65th IEEE Symposium on Foundations of Computer Science (FOCS)*, Oct. 2024.
- [7] **J. Lee**, A. Wibisono, and M. Zampetakis, “Learning exponential families from truncated samples”, in *Advances in Neural Information Processing Systems (NeurIPS)*, Dec. 2023.
- [8] **J. Lee**, S. Haghighatshoar, and A. Karbasi, “Exact gradient computation for spiking neural networks through forward propagation”, in *International Conference on Artificial Intelligence and Statistics, (AISTATS)*, Apr. 2023.
- [9] M. Zhang, **J. Lee**, and S. Agarwal, “Learning from noisy labels with no change to the training process”, in *International Conference on Machine Learning (ICML)*, Jul. 2021.
- [10] K. Jaidka, S. C. Guntuku, **J. H. Lee**, Z. Luo, A. Buffone, and L. H. Ungar, “The rural–urban stress divide: Obtaining geographical insights through twitter”, *Computers in Human Behavior*, vol. 114, p. 106544, Jan. 2021, ISSN: 0747-5632.
- [11] S. Chen, E. Dobriban, and **J. H. Lee**, “A group-theoretic framework for data augmentation”, *Journal of Machine Learning Research (JMLR)*, vol. 21, no. 245, pp. 1–71, 2020.
- [12] S. Chen, E. Dobriban, and **J. Lee**, “A group-theoretic framework for data augmentation”, in *Advances in Neural Information Processing Systems (NeurIPS)*, **Oral Presentation**, 2020.

TECHNICAL REPORTS

- [13] S. Mayhew, T. Tsygankova, F. Marini, Z. Wang, **J. Lee**, X. Yu, X. Fu, W. Shi, Z. Zhao, W. Yin, K. K. J. Hay, M. Shur, J. Sheffield, and D. Roth, “University of Pennsylvania LoReHLT 2019 Submission”, Tech. Rep., 2019.

PROFESSIONAL EXPERIENCE

Google

Student Researcher, CoreML (Domain Applied ML) (Internship/Part-Time)

- Studying solutions for improving performance of LLMs in multi-step reasoning with long contexts.

Mountain View, CA

Summer/Fall 2026

- Meta** Menlo Park, CA
Research Scientist, Video Recommendations (Internship) Summer/Fall 2025
- Studied algorithms for improving video recommendation model using user interest history.
 - Developed methods to process noisy signal (short watch time) videos using sub-sequence-aware unbiased loss (SUL) as an auxiliary loss. Also contributed to work on Virtual Sequential Target Attention (VISTA) model.
- Morgan Stanley** New York, NY
Machine Learning Research Associate (Internship) Summer 2024
- Studied algorithms for solving variational forms of conditional moment problems with applications to regression with heteroskedasticity.
 - Implemented practical solutions on muni bond trading data while analyzing guarantees of algorithms for solving the variational method of moments problem.
- Twitter** San Francisco, CA
Machine Learning Engineer II, Ads Targeting and Modeling (Full-Time) Jul 2019-Aug 2021
- Built ML models and pipelines to help advertisers find their audience on Twitter.
 - Worked fully end-to-end on ML pipelines: from data processing, data pipeline, model architecture and design, training, and serving, on a variety of targeting products including demographic targeting, mobile app recommendation systems, and early filtering models.
- Goldman Sachs** New York, NY
Investment Management Summer Analyst (Private Wealth Management Strats) (Internship) Summer 2018
- Collaborated with portfolio managers/traders to analyze portfolios, create investment algorithms, and build pricing and other models.
 - Built a web tool for managing PWM (Private Wealth Management) clients' preferred stock portfolios. Designed machine learning models to predict proportion of private wealth clients likely to need margin/bank loans to grow the business.
- Morgan Stanley** New York, NY
Quantitative Finance Summer Analyst (Securitized Products Group Strats) (Internship) Summer 2017
- Worked with traders and data to advise new trading strategies in securitized products.
 - Found discrepancies in real loan data to find trading opportunity in RMBS (residential mortgage-backed securities) and used machine learning models to predict loan term modification rate of Freddie Mac loans to find opportunity in CRT (credit risk transfer) products.

FELLOWSHIPS AND STRUCTURED PROGRAMS

- AI Societal Impact Lab** Remote
Inaugural Autumn 2026 Fellow Fall 2026
- Studying the socio-political implications of emerging AI technology.
 - Selected for final cohort of 16 out of 10,000 applicants.
- Mass Atrocities in the Digital Era (MADE)** New Haven, CT
MADE Fellow 2025-2026, 2026-2027
- Studying the role that emerging technology plays in mass atrocities, with a focus on accountability, corporate responsibility, and genocide studies scholarship.

OTHER ACTIVITIES

- 2023-2027 Benjamin Franklin College Graduate Affiliate (Yale)
- 2026 Swiss Winter School on Theoretical Computer Science
- 2025 Women in Theory Workshop at the Simons Institute
- 2023 Citadel PhD Summit

PRESENTATIONS AND INVITED TALKS

- Learning-Enabled Estimation: Tight Characterizations under Sample Selection Biases
 - EC 2026 Main Conference (July 10, 2026)
- Massive Memorization with Hundreds of Trillions of Parameters for Sequential Transducer Generative Recommenders
 - ICLR 2026 Main Conference (April 23, 2026)
- Efficient Statistics With Unknown Truncation: Polynomial Time Algorithms Beyond Gaussians
 - Yale Data Science x Astronomy-Astrophysics Seminar (March 25, 2025) as *Distribution Learning and Inference using Incomplete Data*
 - FOCS 2024 Main Conference (October 19, 2024)
- Learning Exponential Families from Truncated Samples
 - Morgan Stanley Research Group Presentation (March 19, 2024)
 - NeurIPS 2023 Main Conference (December 13, 2023)
 - ICML 2023 2nd ICML Workshop on New Frontiers in Adversarial Machine Learning (July 28, 2023)
 - Yale Theory Student Seminar (June 7, 2023)
- Risk-Averse Constrained Reinforcement Learning with Optimized Certainty Equivalents
 - ICML 2023 Duality Principles for Modern Machine Learning Workshop (July 29, 2023)
 - Citadel PhD Summit (April 14, 2023)
- Exact gradient computation for spiking neural networks
 - AISTATS 2023 Main Conference (April 26, 2023)
 - NeurIPS 2022 Optimization for Machine Learning Workshop (December 3, 2022)
 - TILOS Student and Postdoc Workshop (March 28, 2022)
- A group-theoretic framework for data augmentations
 - UCSD Rose Yu’s research group weekly meeting (May 16, 2022)
 - Workshop for Equivariance and Data Augmentation (September 4, 2020)

TEACHING

I received the Yale Computer Science Department Distinguished Teaching Award for 2022-23.

- **Teaching Fellow** at Yale University Spring 2023
Probabilistic Machine Learning (CPSC 586)
- **Teaching Fellow** at Yale University Fall 2022
Introduction to Database Systems (CPSC 537)
- **Teaching Assistant** at University of Pennsylvania Spring 2018, Spring 2019
Machine Learning (CIS 520)
- **Teaching Assistant** at University of Pennsylvania Spring 2019
Algorithms (CIS 320)
- **Teaching Assistant** at University of Pennsylvania Fall 2018
Agent-Based Modeling and Simulation (ESE 520)
- **Teaching Assistant** at University of Pennsylvania Spring 2018
Internet and Web Systems (CIS 555)
- **Head Teaching Assistant** at University of Pennsylvania Fall 2017
Software Engineering (CIS 573)
- **Teaching Assistant** at University of Pennsylvania Spring 2017
Software Engineering (CIS 350)
- **Head Teaching Assistant** at University of Pennsylvania Fall 2016, Spring 2017, Fall 2017
Data Structures and Algorithms (CIS 121)

SKILLS

- **Technical Skills:** Proficient: Python (Tensorflow, PyTorch), Java; Basic: Scala, MATLAB, SQL, C/C++
- **Language Skills:** Native: English; Conversational: Korean

SERVICE

- **Program Committee**

I have served on the program committee for: ACM Conference on Economics and Computation (EC) 2026

- **Reviewer**

I have served as a reviewer for the following:

- **Conferences:** Innovations in Theoretical Computer Science (ITCS) 2023, IEEE Symposium on Foundations of Computer Science (FOCS) 2024, Conference on Learning Theory (COLT) 2025, Conference on Neural Information Processing Systems (NeurIPS) 2025, International Conference on Learning Representations (ICLR) 2026, ACM Conference on Economics and Computation (EC) 2026, Conference on Neural Information Processing Systems (NeurIPS) 2026, ACM-SIAM Symposium on Discrete Algorithms (SODA) 2027
- **Journals:** Journal of Machine Learning Research (JMLR)
- **Workshops:** Reliable ML from Unreliable Data Workshop (NeurIPS 2025), AXIOM: Foundations of Efficient Deep Learning (NeurIPS 2026), PALM: Personalized, aligned, long-term memory for AI systems (NeurIPS 2026)
- **Graduate and Professional Student Senate (GPSS)** 2024–2025
I served as an elected member of the Graduate and Professional Student Senate (GPSS) for the 2024-2025 school year.
- **Graduate Student Assembly (GSA)** 2022–2025
I served as an elected member on the Graduate Student Assembly (GSA) as a representative of the Computer Science department for the 2022-2023, 2023-2024, and 2024-2025 school years.
 - **Service Committee Chair**, 2022-2023, 2023-2024
 - I served as the elected chair of the Service Committee for the 2022-2023 and 2023-2024 school years, focused on improving service to graduate students and New Haven residents at large.
- **Departmental Graduate Student Advisory Committee (GSAC)** 2022–2024
I served as one of the first members of the newly formed Yale Computer Science Department's Graduate Student Advisory Committee for the 2022-2023 school year. I continued to serve as an elected member for 2023-2024.
- **Ackerman Teaching Award Student Review Committee** 2022–2023, 2024–2025
I was nominated by my department to serve on Yale's School of Engineering and Applied Science (SEAS) graduate student committee to review nominations for the Ackerman Teaching Award for 2022-2023 and 2024-2025.

VOLUNTEERING AND OUTREACH

- Graduate Student Mentor at **Yale Graduate Society of Women Engineers (SWE)** 2023–2025
Volunteer to mentor an undergraduate student who identifies as a woman in engineering. I have committed to make time for advice, support, and meeting throughout the school year.
- Graduate Student Mentor at **Women in Science at Yale (WISAY)** 2021–2025
Volunteer to mentor an undergraduate student who identifies as a woman in science. I have committed to make time for advice, support, and meeting regularly throughout the school year.
- Volunteer Teaching Assistant at **Microsoft TEALS Program** 2021–2024
Volunteer 2-3 days a week to assist AP Computer Science A class at local high school. The TEALS program provides high school students with equitable access to computer science (CS) education and create a pathway to economic opportunity.
- Graduate Student Mentor at **STEM Mentors at Yale** 2021–2024
Volunteer through annual events (2-part college essay writing workshops and career panel) to motivate and support local students grades 6-12 in STEM, especially underrepresented groups like women and minorities.
- Volunteer Tutor at **The SMART Program** 2019–2021

Volunteer weekly to help tutor a high school student in mathematics, ranging from Algebra II + Trigonometry to Precalculus. The SMART program supports low-income middle and high school students in the San Francisco area to break the cycle of poverty.

- Alumna Mentor at **WiCS Alumni Mentorship Program (Penn)** Spring 2020
Volunteered to be an alumni mentor for WiCS (Women in Computer Science) pilot program that aims to connect undergraduate upperclassmen with UPenn Alumni members who also studied CIS/Engineering. Made time for questions, advice, chatting for undergraduate students.